

PREDICTING E-COMMERCE PURCHASE LIKELIHOOD

Feature Importance Analysis Report

Derived from: purchase_prediction_analysis.ipynb

Importance source: Tuned Random Forest classifier (GridSearchCV)

Top 10 Most Influential Features

Executive Summary

This report presents a detailed feature-importance analysis for the e-commerce purchase-prediction project, identifying which of the 12 available customer attributes most strongly influence whether a customer completes a purchase. Feature importances were extracted from the tuned Random Forest classifier's internal impurity-reduction scores, which quantify how much each feature contributed to the model's decision-making across all trees in the ensemble.

The analysis found that purchase behavior signals — specifically prior purchase count and current cart activity — overwhelmingly dominate the model's predictions, together accounting for roughly 70% of total feature importance. Session engagement metrics (time on site, pages viewed) contribute meaningfully but secondarily, at roughly 22% combined. Pricing signals (discount usage, average order value) and demographic or channel attributes (age, gender, traffic source) contribute only marginally, together accounting for less than 8% of the model's decision-making.

This report ranks and explains the top 10 features in detail, cross-references them against the exploratory data analysis (EDA) performed in the notebook, discusses the methodology and limitations of the importance calculation, and closes with recommendations for how these findings should inform both future feature engineering and business strategy.

1. Methodology for Feature Importance Calculation

Feature importance in this analysis was derived from the tuned Random Forest classifier (selected hyperparameters: `max_depth=5`, `max_features='log2'`, `min_samples_leaf=2`, `min_samples_split=5`, `n_estimators=200`), using scikit-learn's built-in impurity-based importance measure. This measure sums, for each feature, the total reduction in Gini impurity it produces whenever it is used to split a node, averaged across all 200 trees in the forest and normalized so that all importances sum to 1.0.

Before importance could be computed, all input features passed through the same preprocessing pipeline used for model training: numeric features (Age, PagesViewed, TimeOnSite, CartItems, PreviousPurchases, AverageOrderValue, DiscountUsed) were mean-imputed and standardized, while categorical features (Gender, DeviceType, TrafficSource) were most-frequent-imputed and one-hot encoded. One-hot encoding expands each categorical feature into one binary column per category — for example, TrafficSource becomes four separate columns (Email, Social Media, Search Engine, Direct) — which is why several of the individual entries in the top-10 list represent specific categories of a single original variable (e.g., "Gender: Female" and "Gender: Male" both appear separately) rather than the variable as a whole.

1.1 Why Random Forest Importance Was Used

Random Forest importance was chosen for this analysis over alternative approaches (such as Logistic Regression coefficients or permutation importance) because it is readily available as a direct output of the tree ensemble, requires no additional computation, and is generally robust to correlated features and non-linear relationships. It is worth noting, however, that the production model ultimately selected for deployment was the optimized Logistic Regression, not the Random Forest — a distinction discussed further in Section 6.

1.2 A Note on Interpreting Impurity-Based Importance

Impurity-based importance measures how useful a feature was for splitting data during training, which correlates with predictive power but is not identical to it. Impurity-based importance can be inflated for

high-cardinality or continuous features (which have more possible split points) relative to binary or low-cardinality features, so caution should be used when comparing a continuous feature like PreviousPurchases directly against a binary one-hot-encoded feature like Gender: Female on the same numeric scale. This is discussed further in Section 7.

2. Top 10 Most Influential Features

The table below lists the 10 highest-ranked features by Random Forest importance score, along with the broad category of customer information each feature represents.

Rank	Feature	Importance	Category
1	PreviousPurchases	0.4623 (46.2%)	Purchase history
2	CartItems	0.2431 (24.3%)	Current-session activity
3	TimeOnSite	0.1407 (14.1%)	Engagement
4	PagesViewed	0.0801 (8.0%)	Engagement
5	DiscountUsed	0.0311 (3.1%)	Pricing / promotion
6	AverageOrderValue	0.0153 (1.5%)	Spending history
7	Age	0.0126 (1.3%)	Demographic
8	TrafficSource: Search Engine	0.0025 (0.3%)	Acquisition channel
9	Gender: Female	0.0020 (0.2%)	Demographic
10	Gender: Male	0.0018 (0.2%)	Demographic

Two features — PreviousPurchases and CartItems — together account for approximately 70.5% of total model importance, making them by a wide margin the most decision-relevant inputs available to the model. The next two — TimeOnSite and PagesViewed — add a further 22.1%, meaning the top four features alone explain over 92% of the Random Forest's total predictive weighting. The remaining six features in the top 10, plus the two features that did not make the list (DeviceType categories and the remaining TrafficSource categories), collectively account for less than 8% of importance.

3. Deep-Dive Analysis of Top Features

3.1 PreviousPurchases (Importance: 0.4623 — Rank 1)

A customer's count of previous completed purchases, ranging from 0 to 9 in the dataset, is the single most important input to the model by a wide margin — nearly double the importance of the next-ranked feature. This aligns strongly with well-established e-commerce intuition: customers who have purchased before have demonstrated trust in the platform, familiarity with the checkout process, and (often) an established product preference, all of which make repeat purchase substantially more predictable than a first-time conversion. This feature's dominance suggests that the model is, in large part, a sophisticated way of recognizing "this is a customer who buys" versus "this is a customer who is still deciding whether to buy at all."

From a modeling perspective, this also means that any customer with no purchase history (a true first-time visitor) is inherently harder for this model to classify with confidence, since the model's strongest signal is unavailable for that customer by definition. This has direct implications for how the business should treat new-visitor conversion, discussed in Section 5.

3.2 CartItems (Importance: 0.2431 — Rank 2)

The number of items currently in a customer's cart, ranging from 0 to 5, is the second most influential feature. Unlike PreviousPurchases, which reflects long-term history, CartItems is a real-time, within-session signal — it captures active purchase intent as it is happening. The notebook's exploratory boxplot analysis (comparing CartItems between customers who purchased and those who did not) shows a visibly higher median and interquartile range of cart items among purchasers, confirming that this feature has genuine, visually apparent separating power, not just statistical importance inside the model.

Because this signal updates during a live session, it is also the most actionable feature for real-time intervention: a business system could monitor CartItems as it changes and trigger different treatments (reminder banners, incentives, simplified checkout flows) as a customer's cart grows, rather than waiting until the session ends to act.

3.3 TimeOnSite (Importance: 0.1407 — Rank 3)

Time spent on the site, measured in minutes and ranging from 1 to 9 in this dataset, is the third most important feature. The notebook's boxplot comparison shows customers who purchased tend to spend more time on site than those who did not, though with more overlap between the two groups than is seen for CartItems — consistent with its lower importance score. Time on site is best understood as a general engagement signal: it indicates a customer is actively considering the site's offerings, but by itself does not confirm intent to buy as strongly as adding items to a cart does.

3.4 PagesViewed (Importance: 0.0801 — Rank 4)

The number of distinct pages viewed during a session, ranging from 1 to 14, ranks fourth. Like TimeOnSite, this is an engagement-depth signal, capturing how thoroughly a customer is browsing the catalog. Its lower importance relative to TimeOnSite suggests that how long a customer spends matters somewhat more to the model than how many pages they click through — plausibly because a customer who lingers on a small number of product pages may be evaluating a specific purchase decision, while rapid page-flipping alone does not guarantee genuine interest.

3.5 DiscountUsed (Importance: 0.0311 — Rank 5)

Whether a customer applied a discount code (a binary 0/1 flag) ranks fifth, with a modest but non-trivial importance of about 3.1%. The notebook's crosstab and bar-chart analysis show a visible lift in purchase rate among customers who used a discount, confirming the direction of the effect (discounts are associated with higher purchase rates) even though its overall contribution to the model is small relative to behavioral signals. This suggests discounts have a real but limited independent effect once purchase history and cart activity are already accounted for — in other words, much of the apparent "discount effect" seen in raw crosstabs may be partly explained by the fact that highly engaged, high-intent customers are also more likely to seek out and apply discount codes in the first place.

3.6 AverageOrderValue (Importance: 0.0153 — Rank 6)

The customer's historical average order value, ranging from 20 to 299 currency units, ranks sixth with roughly 1.5% importance. This is a comparatively weak standalone predictor of whether the current session

will convert — a customer with a high historical average order value is not meaningfully more or less likely to complete a new purchase than a lower-value customer, based on this model. This feature may be more useful for a related but different business question (e.g., predicting the size of an order, rather than whether one occurs), and is discussed further in Section 6.

3.7 Age (Importance: 0.0126 — Rank 7)

Customer age, ranging from 18 to 59 in the dataset, ranks seventh with about 1.3% importance — a weak predictor of purchase likelihood on its own. This suggests purchase decisions in this dataset are driven far more by what a customer is doing (behavior) than by who they are (demographics), a pattern consistent with the very low importance scores for the remaining demographic and channel features.

3.8 TrafficSource: Search Engine (Importance: 0.0025 — Rank 8)

Among the four acquisition channels captured in the data (Email, Social Media, Search Engine, Direct), the Search Engine category was the only one to place in the top 10, with a small importance of about 0.25%. This suggests customers arriving via search may carry marginally higher or more distinguishable purchase intent than those from other channels (plausibly because search traffic often reflects an active, specific product query rather than passive discovery), but the effect is small enough that acquisition channel should not be treated as a meaningful standalone predictor of conversion.

3.9–3.10 Gender: Female and Gender: Male (Importance: 0.0020 and 0.0018 — Ranks 9–10)

The two gender categories round out the top 10 with negligible importance (0.20% and 0.18% respectively). Because one-hot encoding represents a single two-category variable (Gender) as two separate binary columns, these two entries are best interpreted together as a single finding: gender, as captured in this dataset, has essentially no independent bearing on purchase likelihood once behavioral features are accounted for.